Inflammatory Stress Contributes to Tau Pathology in the PS19 Mouse Model of Alzheimer's Disease

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Introduction

Alzheimer's disease (AD) is a complex neurodegenerative disease that causes behavioral deficits and pathophysiological hallmarks, including, tau and amyloid-β dysfunction and accumulation, neuroinflammation and subsequent synaptic loss and neuronal death. Neuroinflammation contributes to tau neurofibrillary tangle (NFT) formation (1). Microglia phagocytose debris and astrocytes form the blood brain barrier (2). The type, severity, and duration of inflammation contribute to neurodegeneration (3). The goal of this study is to determine the relationship and function between peripheral chronic inflammatory stress and behavioral and pathological hallmarks of AD—tau accumulation and immune system activation. We hypothesize that inflammatory stress increases pathological tau burden and contributes to the pro-inflammatory state characteristic of AD.

Methods

This study used PS19 mice—harboring a microtubule associated protein tau (MAPT) mutation (P301S). Non-Tg (C3H and C57BL/6) littermate counterparts were used as controls.

Mice received intraperitoneal injections (I.P.) of either infection mimetics (I.F.) (pro-inflammatory compounds: Zymosan A, Poly I:C, ODN2395, LPS) or mammalian ringers (control). Injections were administered starting at 6 months of age, spaced 1 month apart per injection, with a total of 4 injections per mouse.

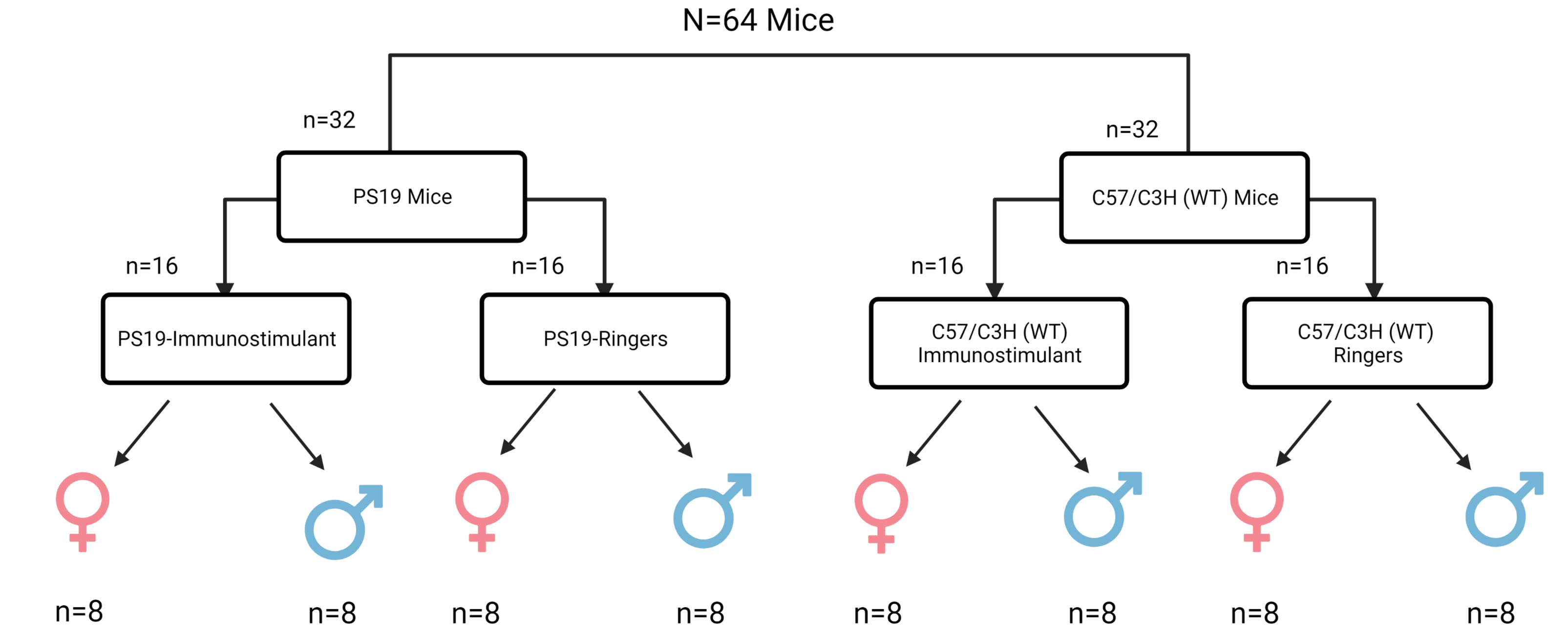


FIGURE 1: Experimental Design

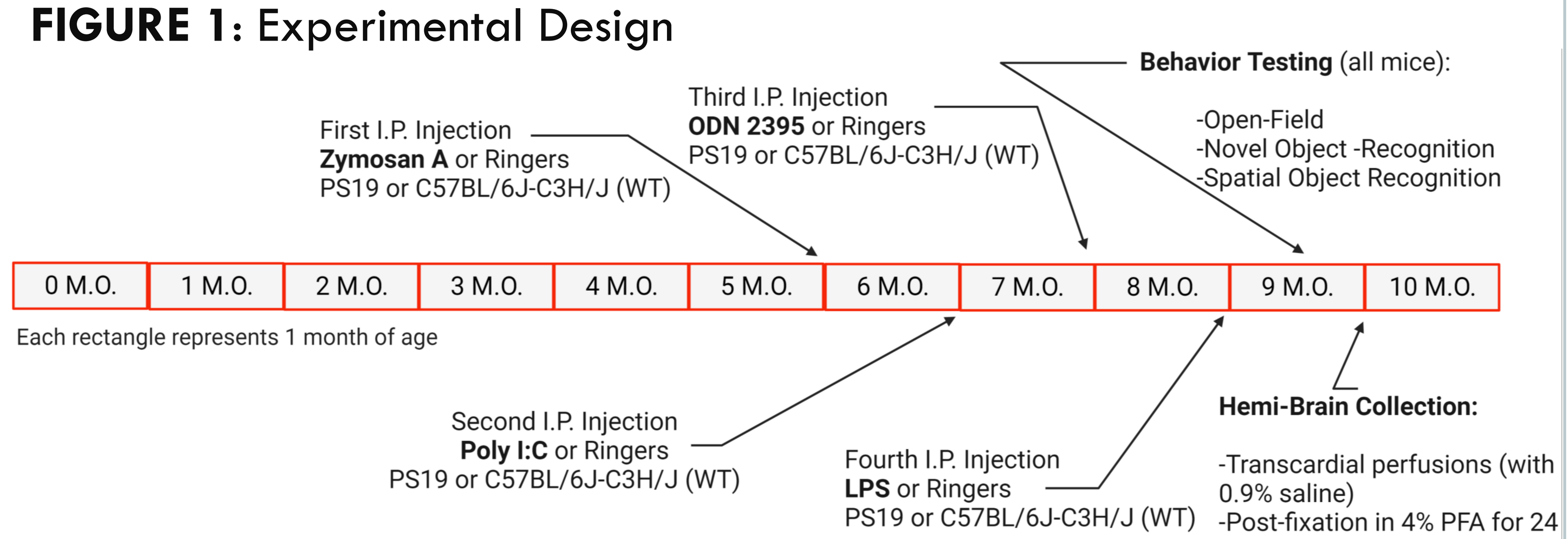


FIGURE 2: Experimental timeline, including I.P. injections, behavioral testing, hemi-brain collections, microscopy, and quantification.

Tau IHC Results-AT8 and HT7

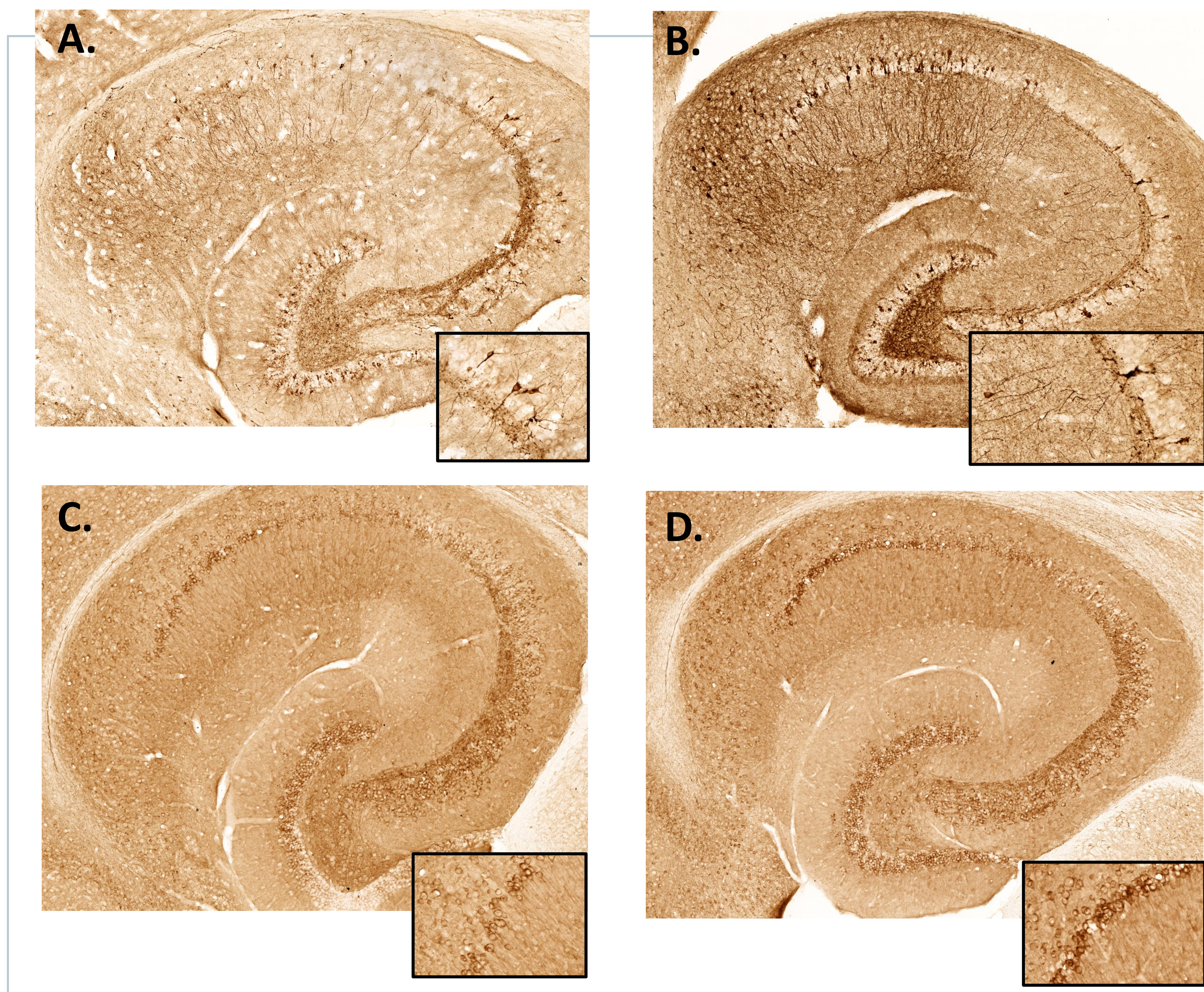
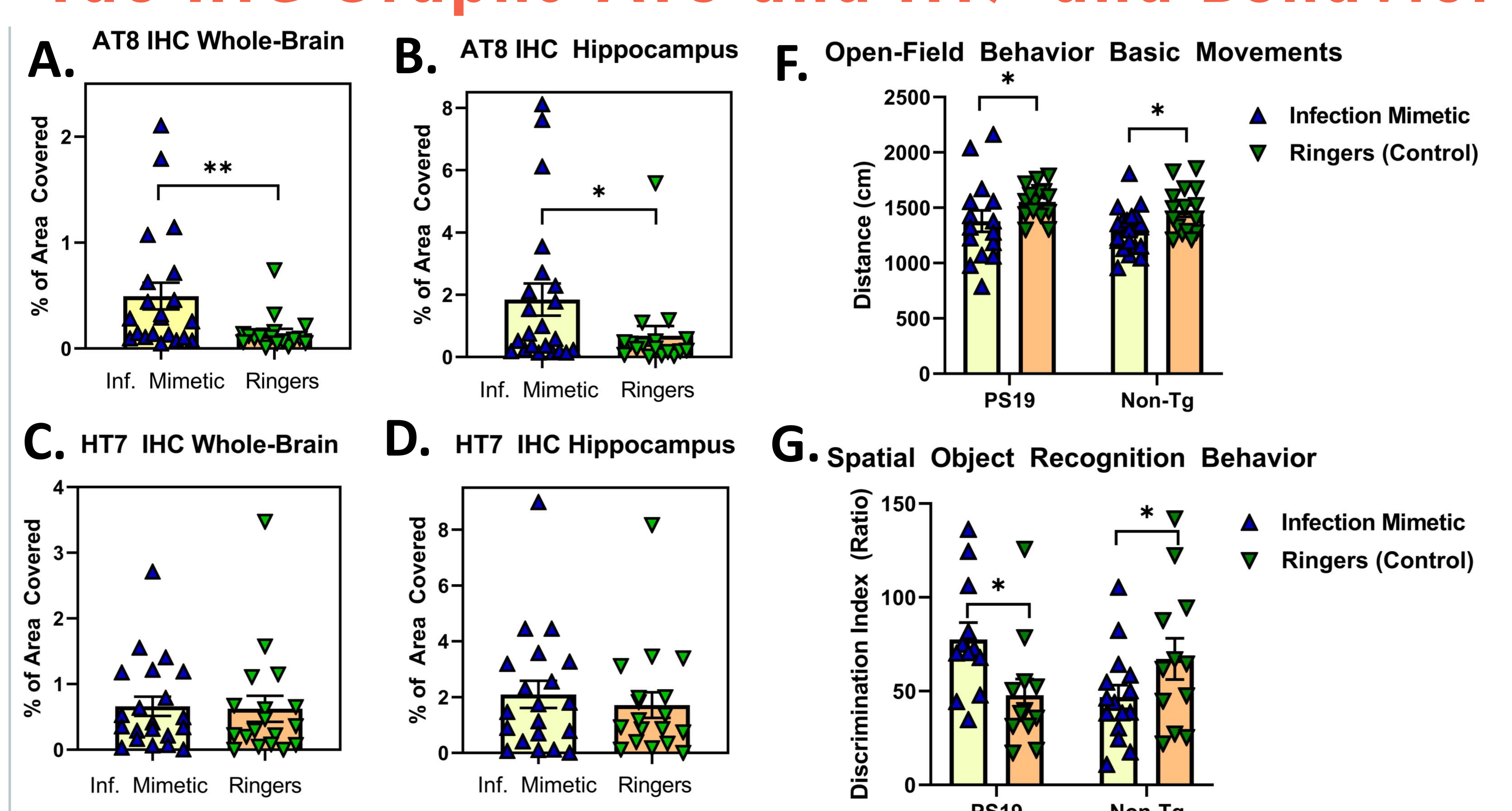


FIGURE 3: Tau IHC results, 20x objective. AT8 (p-tau) **A.** PS19 Ringers, **B.** PS19 I.F. HT7 (total human tau) **C.** PS19 Ringers, and **D.** PS19 I.F.

Tau IHC Graphs-AT8 and HT7 and Behavior



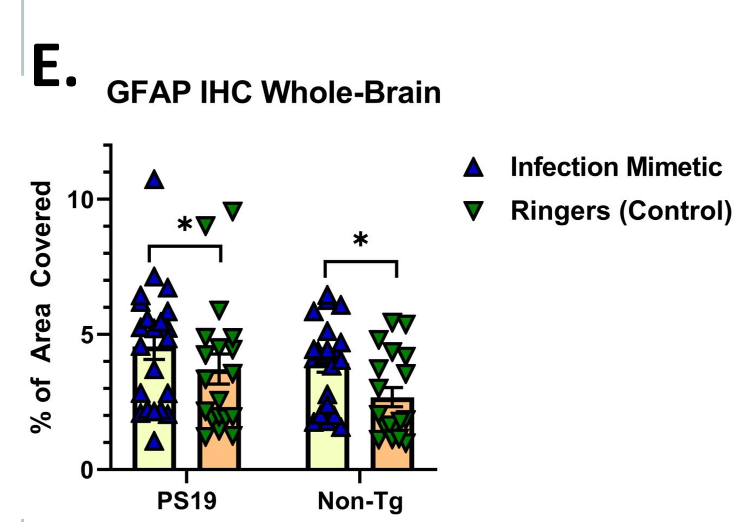


FIGURE 5: AT8 was greater in I.F. mice compared to Ringers in the **A.** whole-hemisphere (WH) ($p = 0.0063$), and in the **B.** hippocampus ($p = 0.049$). Total tau did not sig differ (HT7) **C** in the WH or **D.** the hippocampus. WH GFAP was significantly elevated in the **E.** I.F. group compared to the PS19 Ringers groups ($p = 0.0226$). **F.** I.F. mice showed elevated basic movement ($p = 0.0367$) **G.** and SOR ($P = 0.0173$).

Astrocyte Reactivity (GFAP) IHC Results

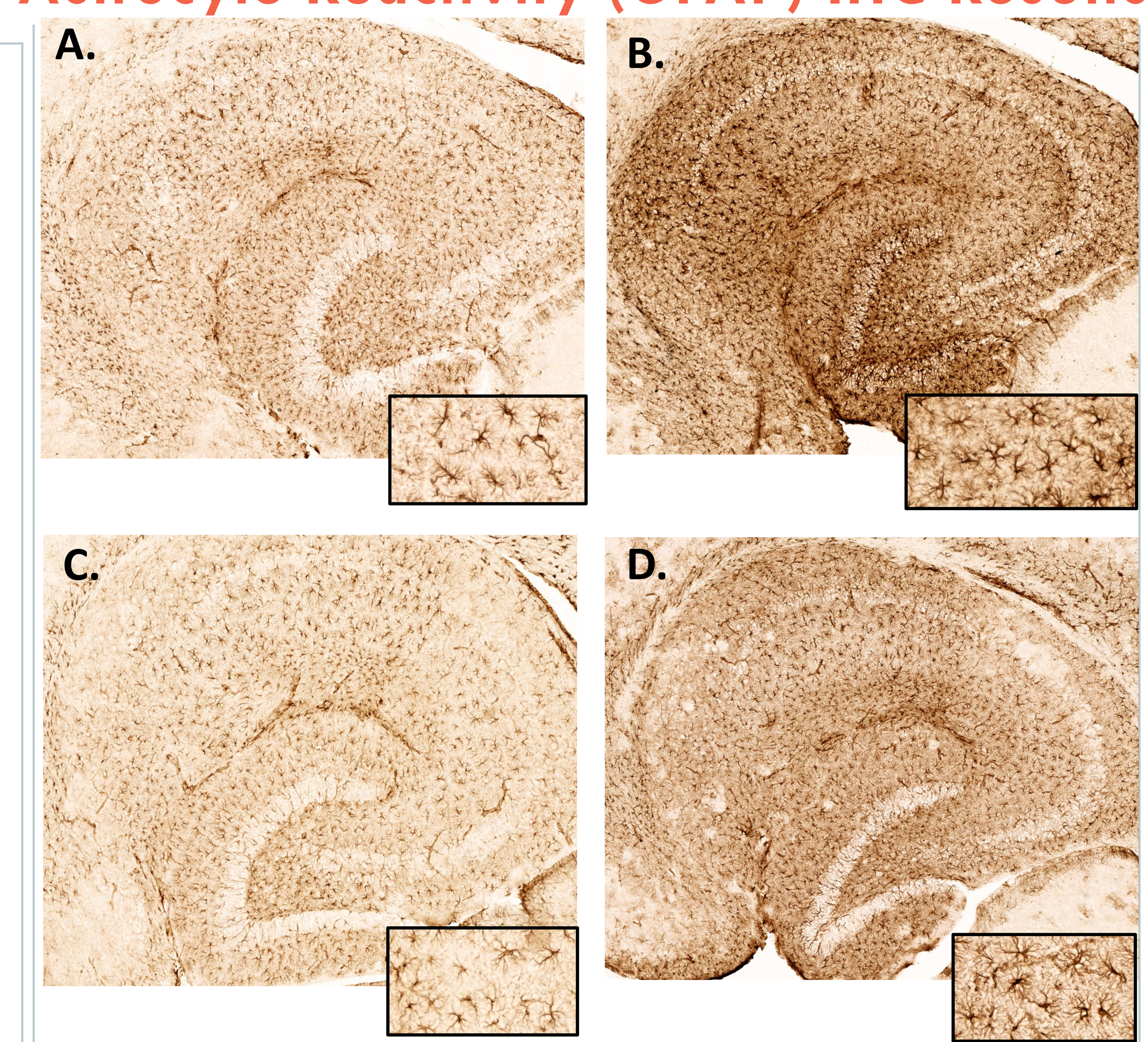


FIGURE 4:-GFAP IHC results, 20x objective. **A.** PS19 Ringers, **B.** PS19 I.F., **C.** Non-Tg Ringers, and **D.** Non-Tg I.F.

Summary

Peripheral and chronic inflammatory stress contribute to tau dysfunction—hyperphosphorylation and aggregation in the PS19 mouse model of AD. Inflammatory stimuli did not significantly alter total tau (HT7), suggesting that the immune system plays a specific role affecting only pathological forms of tau. The pro-inflammatory response may contribute to microtubule dissociation and accumulation in AD. These findings demonstrate that the immune system plays an important role in the onset and progression of the disease.

References

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